

Rehand Abdelhady Yassin

Aerospace Engineer

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Aerospace engineering graduate with practical experience in workshop machines, coding, and teaching. Skilled in CNC machining, Python, C++, and MATLAB. Delivered mathematics instruction at advanced levels. Proficient in SolidWorks, TinkerCAD, Multisim, Proteus, LT Spice, Arduino, Mathematica, and ANSYS. Led a graduation project to design and build a combustion chamber using kerosene, green hydrogen, and their blends.

EDUCATION

- **Zewail City of Science and Technology (Was accepted with Scholarship)** September 2020 - June 2025
Bachelor of Science and Engineering – Aerospace Engineering Giza, Egypt

EXPERIENCE

- **Zewail City of Science and Technology** Present
Undergraduate Internship Mentor 6th October, Giza
 - Supervised and mentored undergraduate students in the design and implementation of a CNC foam cutter
 - Provided hands-on training in stepper motor synchronization, mechanical assembly, and thermal control
- **Egyptian Space Agency (EgSA)** Present
Mechanical Design Engineer-Internship New Cairo City, Cairo
 - Participated in the complete design and configuration of a satellite, including all major subsystems (structural, thermal, power, and payload)
 - Conducted finite element analysis (FEA) on structural components to ensure mechanical integrity under launch and operational conditions
- **Ministry of Youth and Sports** Present
Young Negotiator at EG Youth for Climate Mohandeseen, Giza
 - Representing Egypt as part of the Egyptian Youth Delegation for Climate Action, contributing to national discussions and international negotiations as a part of the Innovation and Solution Committee.
- **Be An Ambassador Initiative** October 2024
Internship Cairo, Egypt
 - Developed and managed the completion of 32 educational videos focusing on sustainability and climate action, enriching a course on climate policies.
 - Completed a one-week intensive training program on soft skills to enhance personal and professional development.
- **Student Energy** March - November 2024
Fellowship Online
 - Completed a fellowship in energy systems and project development, gaining versatile skills for diverse career paths
 - Completed Tier 2 content-focused track on energy transition and sustainability.
- **McKinsey's Forward program** November 2023- April 2024
Training Online
 - Completed 8-week foundation program in adaptability, communication, problem-solving, and digital skills.
 - Completed 11-week advanced program with workshops and projects, enhancing digital expertise and applied skills.
- **Pulse Jet Engine - Zewail City** December 2023
Mechanical Design Engineer Internship 6th October, Giza
 - Orchestrated the development of a classroom demonstration setup for thermofluidic systems based on simplified Brayton and Lenoir cycles. Conducted numerical analysis using MATLAB and utilized SolidWorks and ANSYS for design and analysis.
 - Engineered a cost-effective alternative for aerospace and power generation with a valved pulse jet engine, enhancing fuel efficiency and startup procedures through a universal test rig—managed component procurement and conducted testing in the university workshop environment.
- **EgyptAir Training Academy** September 2023
Maintenance Engineer Internship Cairo, Egypt
 - Exploring Hangar 6000, 7000, and 8000, focusing on gas dynamics, gas turbine engines, and structural repairs.
 - Developing attention to detail and understanding of aircraft maintenance standards, and exploring gas turbine engines, landing gear mechanisms, and Non-Destructive Testing
- **University of Science and Technology at Zewail City** Month Year - Month Year
Junior Teaching Assistant 6th October, Giza
 - Aided Professor and TA in math courses, utilizing my A-grade proficiency to assist students in understanding complex concepts effectively, and developed online and offline tutorials to facilitate student learning and enhance an overall understanding of the course material

PROJECTS

• Dino Runner Game

May 2023

C++, SFML Library

- Developed a game with a graphical user interface using C++ and SFML.
- Implemented object-oriented programming principles to manage game entities and interactions.

• Satellite Design Project

July - August 2025

SolidWorks, ANSYS, MATLAB

- Designed the complete satellite assembly in SolidWorks, integrating structural and subsystem components.
- Performed Finite Element Analysis (FEA) in ANSYS to evaluate stress, deformation, and safety factors.
- Applied mechanism calculations in MATLAB to validate kinematic and dynamic performance.

SKILLS

Mechanical

- ANSYS
- Inventor
- SOLIDWORKS

Programming

- C++
- C
- Python

Soft Skills

- Multitasking
- Animaker
- Prezi

Electrical & Control

- MATLAB & Simulink
- Proteus
- Arduino

HONORS AND AWARDS

• 2nd Place at Graduation Project Fair

June 2025

University of Science and Technology at Zewail City

- Developed a lab-scale combustion chamber using kerosene-green hydrogen blends. Designed the chamber in SolidWorks. Used MATLAB and Simulink for combustion modeling.
- Applied ANSYS to analyze combustion efficiency, pressure, temperature, and density.

• 4th Place at Medhat Ben Zohair Competition

October 2022

Greenish Club

- A competition in entrepreneurship, where the idea works to provide job opportunities for handicraft owners in the desert and rural areas, such as Minya, through tourism work there to learn about customs and traditions

• Global Nominees

October 2022

NASA Space Apps

- Simplified Parker Solar Probe journey using stations on a board, from Earth to the sun's corona layer, integrating QR code educational game using some applications for coding and graphics like G-Develop and Unity

VOLUNTEER EXPERIENCE

• Zewail City Racing Team (Velox Team)

March 2023-June 2024

Ever Competition

- Assign tasks and collect all the designed parts of the wheel assembly and brakes into one assembly.

• Young Professional Program

February 2024

5th settlement, Cairo, Egypt

- Attending the Young Professional Program, which primarily focuses on sustainability, green, and renewable energy technology.

• Be an ambassador Graduation Ceremony

February 2024

Cairo Opera House

- Managed to organize the graduates of this initiative throughout the day, were some ministers coming

• Admission Team

May 2021-Jan 2022

Zewail City

- Facilitated new student onboarding, mentoring in registration, and addressing admission test and procedure queries.

• Future Engineers – NASA

May 2019- June 2023

Online

- Actively participated as a competitor in esteemed competitions such as the "Name the Rover Essay Contest," "Next Moon Step Challenge," and "Reinvent The Rover Wheel Challenge," garnering certificates of participation.
- Currently serving as a competition judge, contributing to the evaluation and adjudication processes.

• Zewail City Science Festival (ZCSF)

June 2021 – August 2023

Zewail City

- Engaged in the event for three consecutive years, serving as an organizational team member in the first year, assuming responsibilities within the HR team in the second year, and contributing as a photographer team member in the third year.

• Zewail City Commencement

July 2023

Zewail City

- I was selected to actively contribute to coordinating and organizing the university's graduation ceremony.